The University of Faisalabad

Computer Organization and Assembly language

Course: Computer Organization and Assembly Language (Lab) - CS-318

Bit Shifting in 8086 Assembly

Shift Instructions

SHL (Shift Left)

- Moves bits to the left
- Fills rightmost bit with 0
- Leftmost bit goes into CF
- Same as multiply by 2 each shift

SHR (Shift Right)

- Moves bits to the right
- Fills leftmost bit with 0
- Rightmost bit goes into CF
- Same as divide by 2 each shift

Shift Left (SHL)

```
org 100h

mov al, 05h       ; 00000101b

shl al, 1         ; becomes 00001010b (= 0Ah)
```

mov ah,4Ch

int 21h

Shift Right (SHR)

org 100h

mov bl, 80h   ; 10000000b

shr bl, 1   ; 01000000b (= 40h)

mov ah,4Ch

int 21h

Multi-Bit Shift Using CL

org 100h

mov al, 0F0h   ; 11110000b

mov cl, 3

shl al, cl   ; shift left 3 times

mov ah,4Ch

int 21h

Shift Left Until Carry (CF) and Jump

ORG 100h

MOV AL, 05h   ; 00000101b

loop1:

SHL AL, 1   ; Shift left by 1

JC end1   ; Jump if Carry Flag = 1 (MSB became 1)

JMP loop1   ; Repeat if CF = 0

end1:

```
MOV AH, 4Ch      ; Terminate program
INT 21h
```

Shift Right Until Zero and Jump

```
ORG 100h
MOV AL, 10h      ; 00010000b
loop2:
SHR AL, 1        ; Shift right by 1
JZ end2          ; Jump if AL = 0
JMP loop2        ; Repeat if AL != 0
end2:
MOV AH, 4Ch      ; Terminate program
INT 21h
```

Lab Exercise

1. Multiply 03h by 4 using **SHL**, then add 02h to the result and store it in AL.
2. Divide 40h by 2 using **SHR**, then compare the result with 15h and jump to a label if it is greater.
3. Shift DL left by 1 and jump to a label if the **Carry Flag (CF)** is set.
4. Shift AH right by the number in CL using **SHR** and jump to a label if the result is zero.
5. Shift AL left by 1. If **CF = 1**, subtract 01h from AL and store the result; otherwise leave AL unchanged.
